Greenhouse Management Design

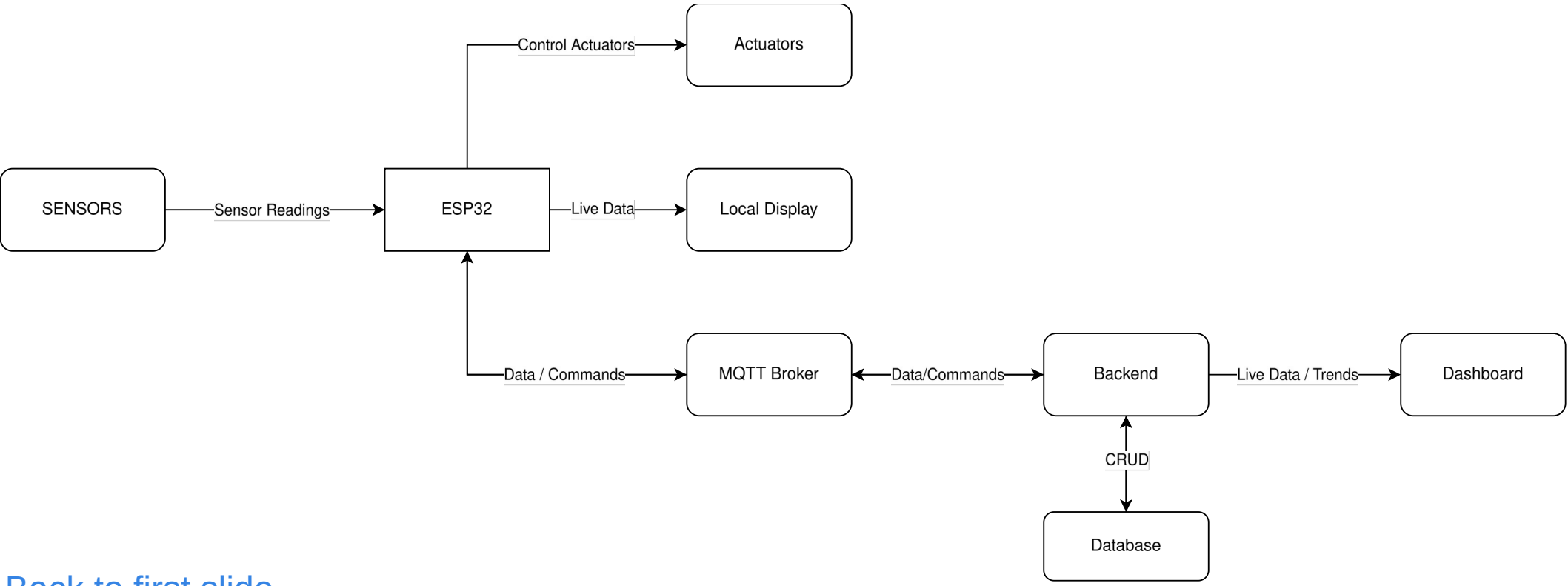
The presentation is separated into 4 parts:

1. Overview
2. Wiring and Schematics
3. Modules
4. Firmware

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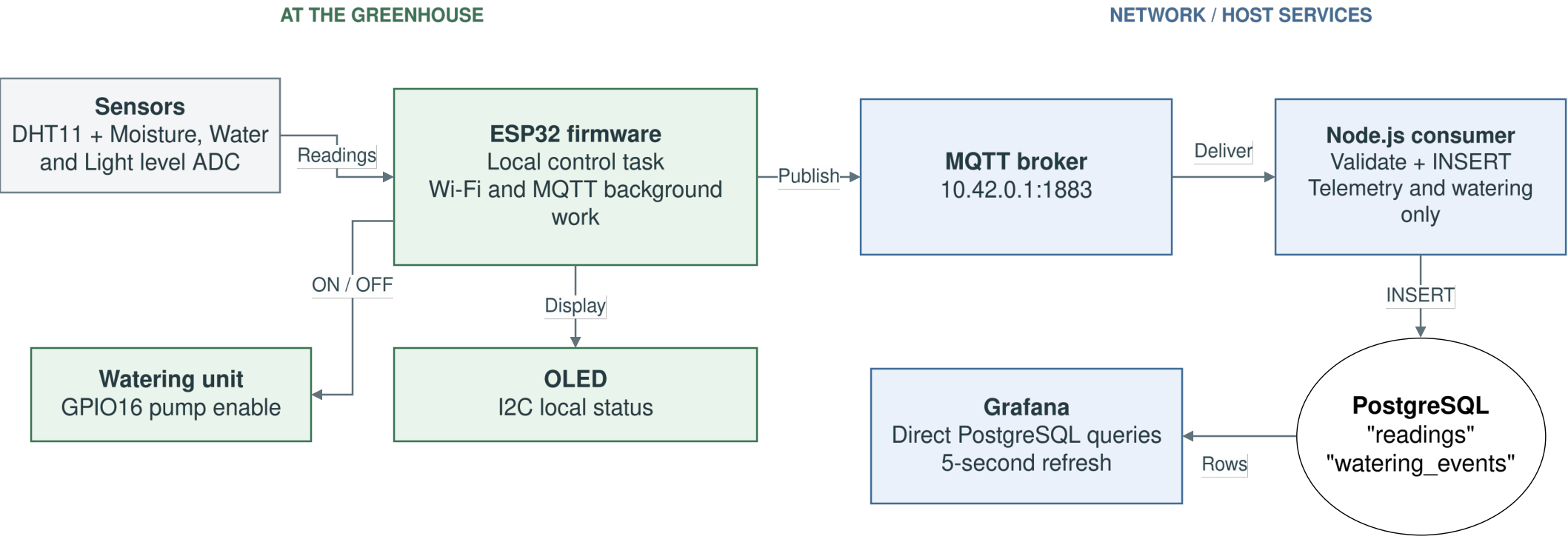
Overview

1.1. System architecture



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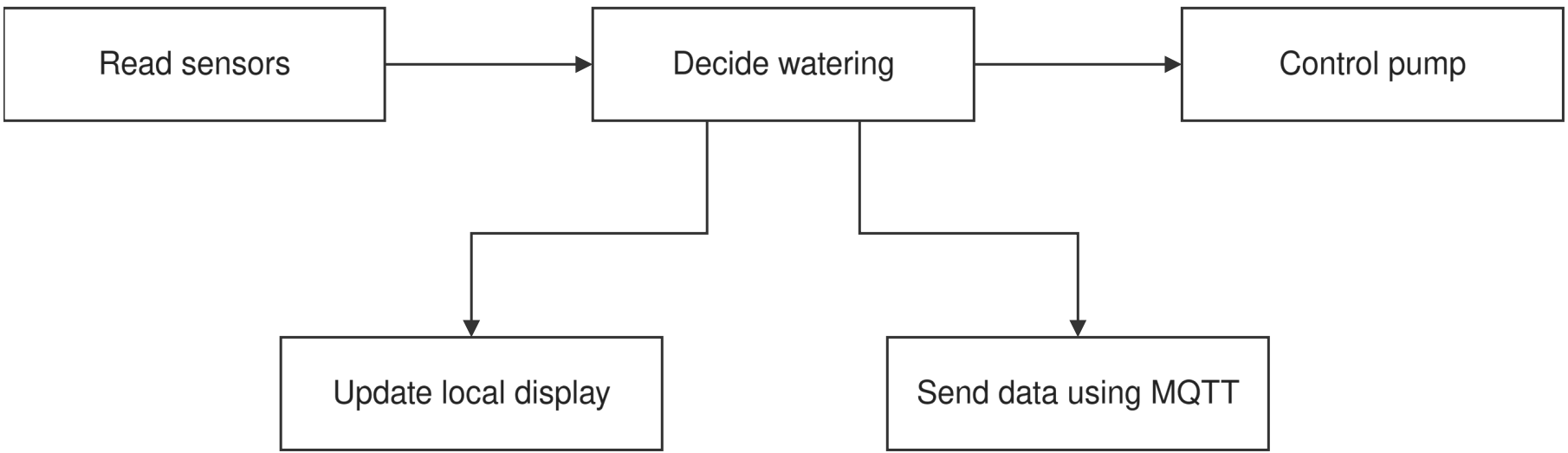
1.1. System architecture - detailed



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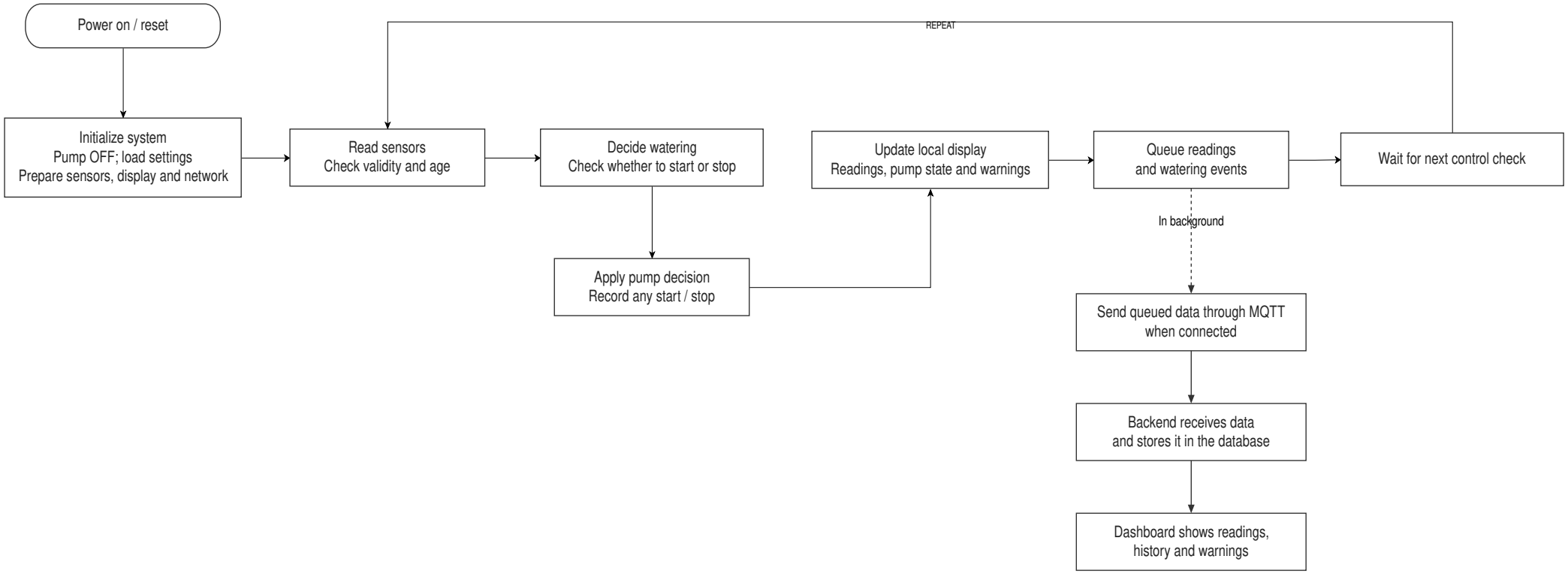
1.2. Firmware responsibilities

ESP32 software



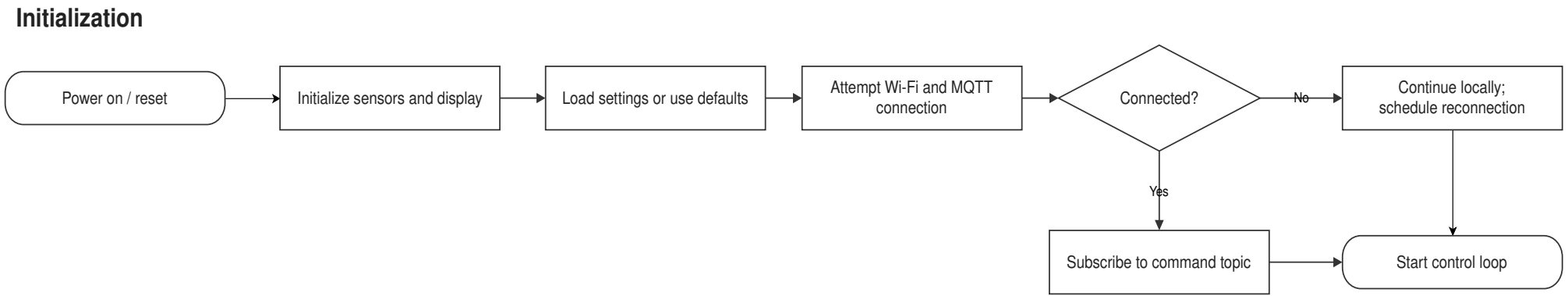
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1.3. Whole system – more detailed



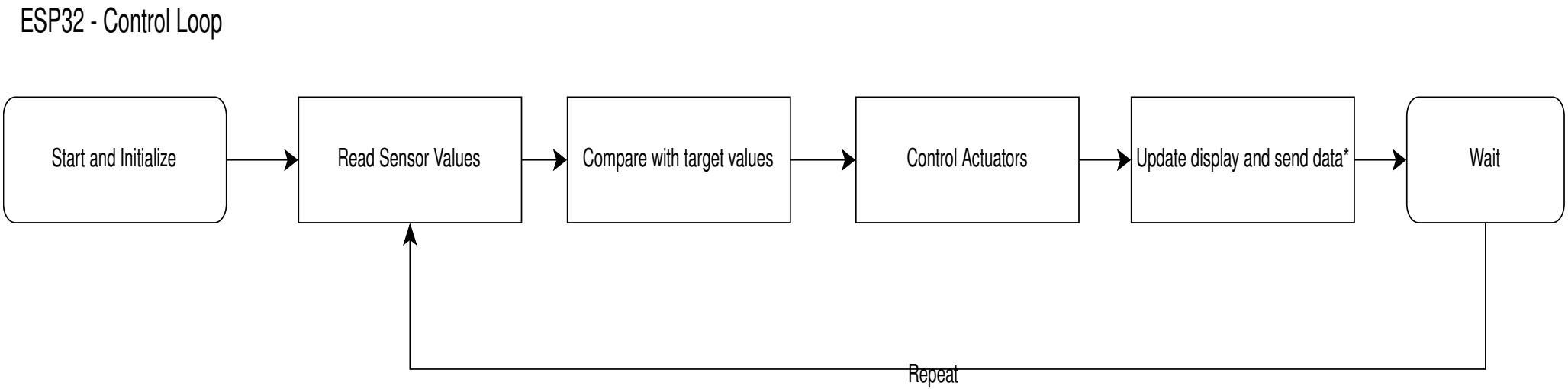
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1.4. Initialization / setup part



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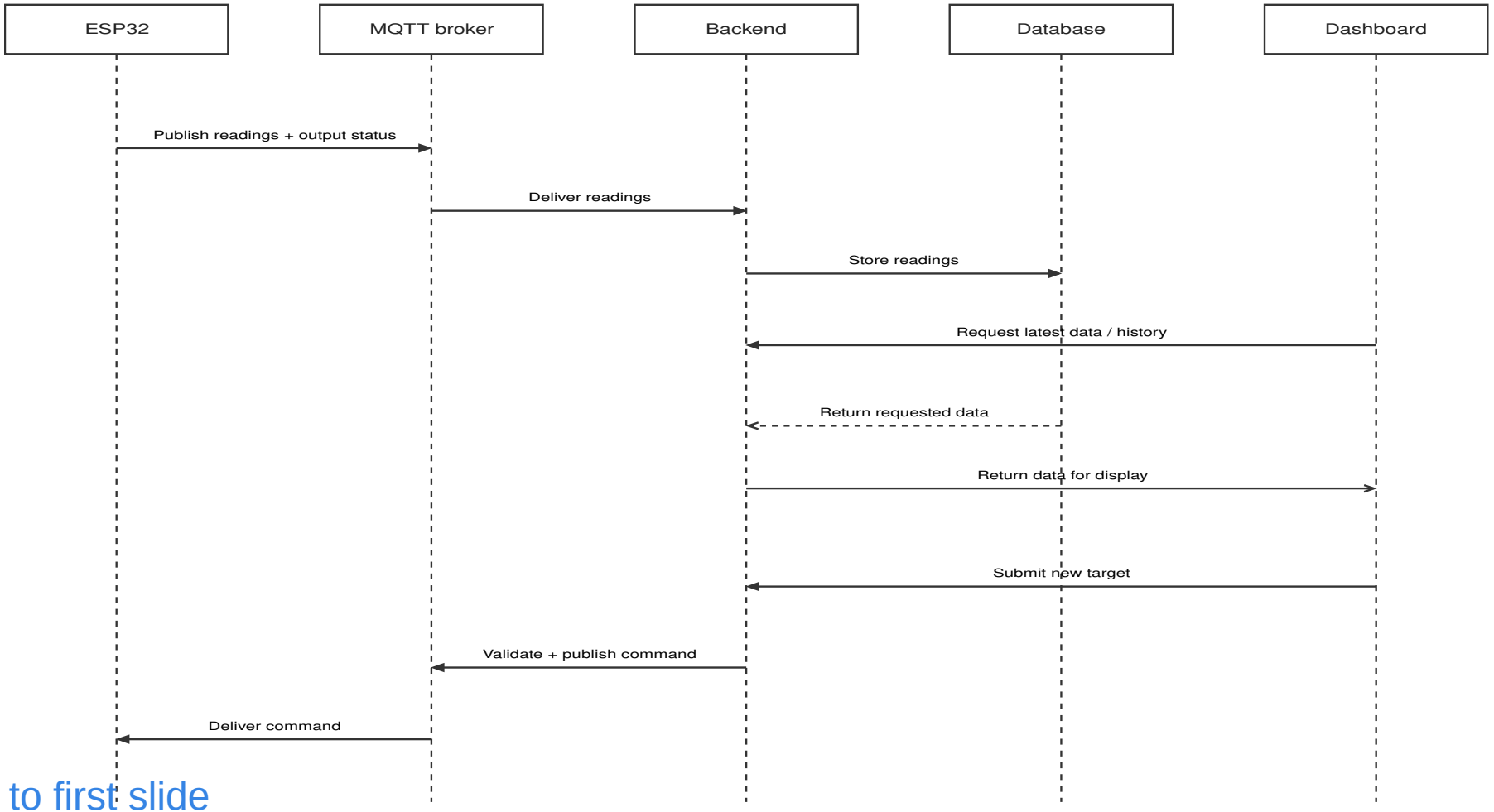
1.5. Control Loop



*Send to the backend through MQTT when connected.

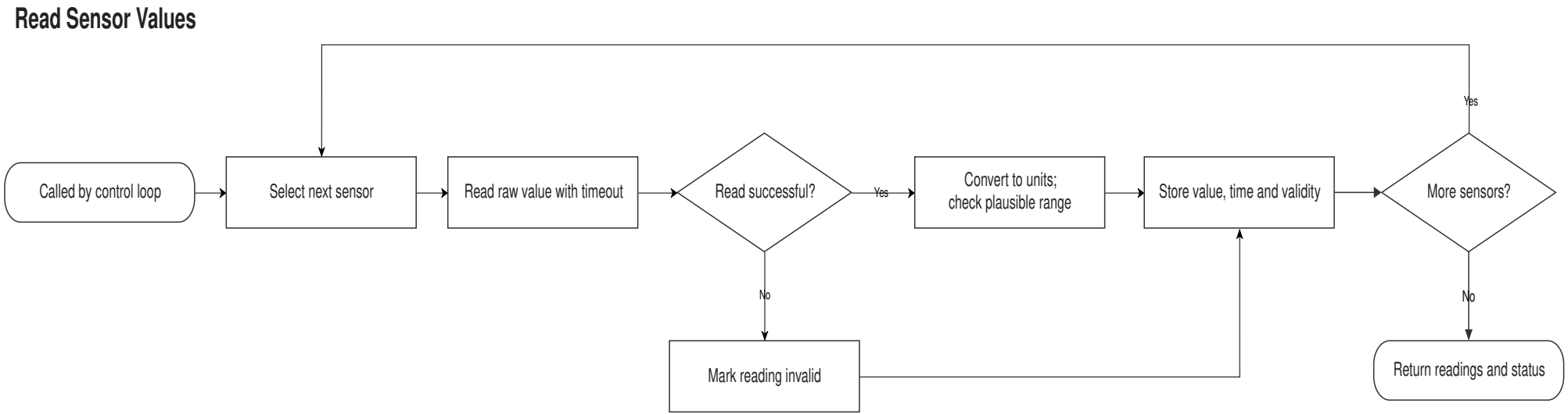
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1.6. Communication protocol



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1.7. Sensor Readings



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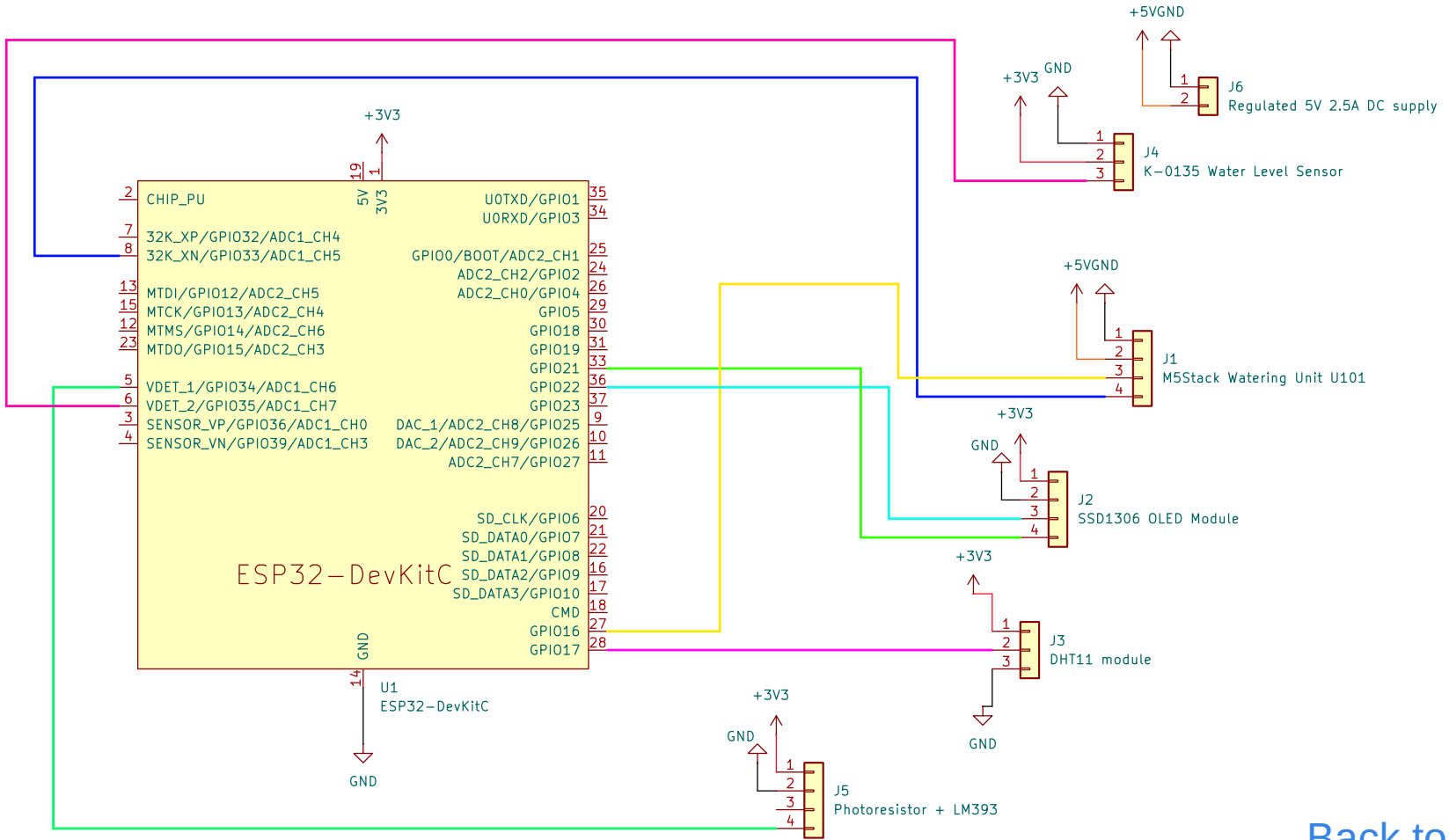
Wiring and Schematics

2.1. Wiring table

Component	**Component signal**	**ESP32 GPIO**	**Interface / behavior**
Soil-moisture sensor	Analog output	GPIO33	ADC1 channel 5
Water-level sensor	Analog output	GPIO35	ADC1 channel 7
Light sensor	Analog output	GPIO34	ADC1 channel 6
DHT11 temperature/humidity sensor	DATA	GPIO17	Digital sensor communication
Pump control	PUMP_EN	GPIO16	HIGH = ON; LOW = OFF
SSD1306 OLED, 128 × 64	SDA	GPIO21	I²C data
SSD1306 OLED, 128 × 64	SCL	GPIO22	I²C clock

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2.2. KiCad Wiring Schematic



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2.2. Power Wiring Table

Component	Power connection	Ground connection
ESP32 DevKitC — U1	USB through Micro-USB socket	ESP32 GND → common GND
External 5V supply — J6	J6 pin 2 (+5V) → J1 pin 2	J6 pin 1 (negative) → common GND
Watering unit: soil sensor + pump — J1	J1 pin 2 → external +5V	J1 pin 1 → common GND
SSD1306 OLED — J2	J2 pin 1 → ESP32 3V3	J2 pin 2 → common GND
DHT11 module — J3	J3 pin 1 → ESP32 3V3	J3 pin 3 → common GND
Water-level sensor — J4	J4 pin 2 → ESP32 3V3	J4 pin 1 → common GND
Light-sensor module — J5	J5 pin 1 → ESP32 3V3	J5 pin 2 → common GND

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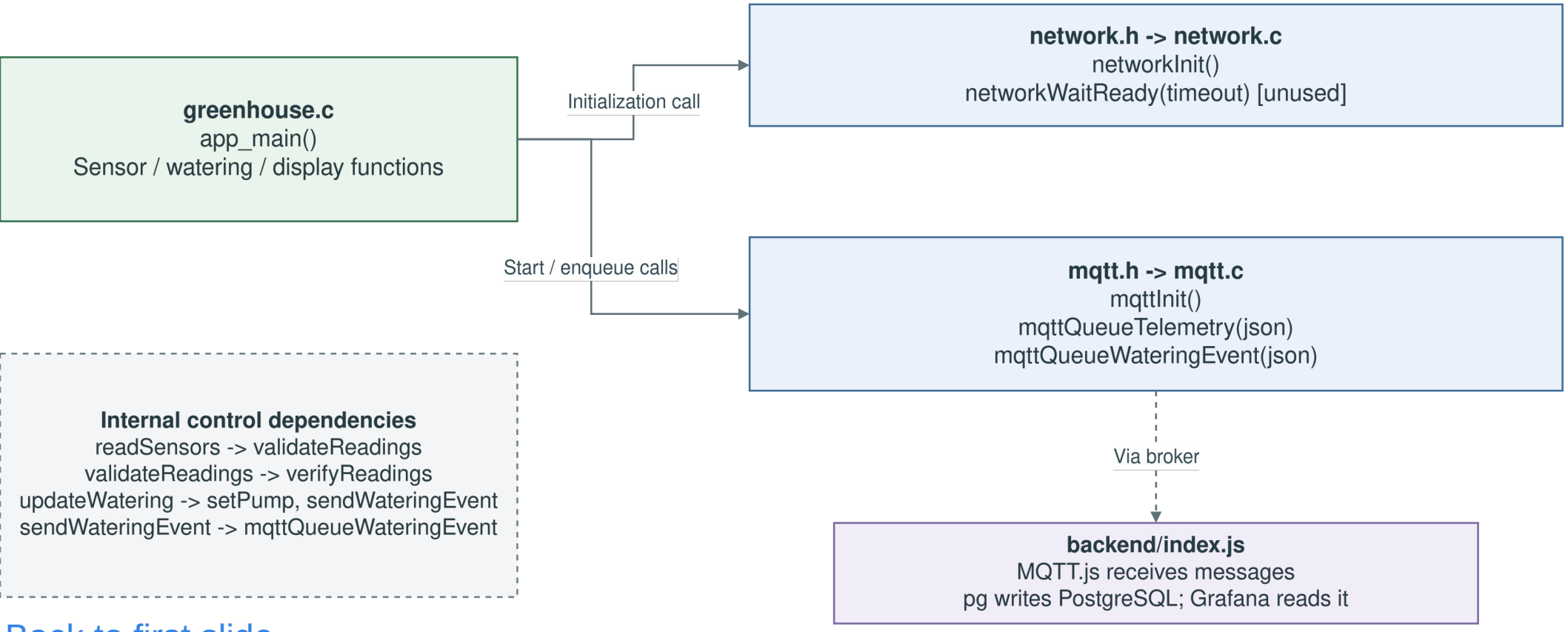
Modules

3.1. Source files and responsibilities

Runtime area	Files	Responsibility
ESP32 application	greenhouse/main/greenhouse.c	Sensors, validation, watering, OLED, payload formatting and main loop
ESP32 networking	greenhouse/main/network.c + network.h greenhouse/main/mqtt.c + mqtt.h	Wi-Fi initialization and retry events MQTT startup, enqueue APIs, status events
Build and hardware libraries	greenhouse/main/CMakeLists.txt idf_component.yml; sdkconfig	Build three C sources; configure ESP-IDF DHT, SSD1306 and MQTT dependencies
Host ingest and transport	backend/index.js; package.json mqtt-test.sh	MQTT.js consumer + pg database inserts Launch broker and diagnostic subscriber
Dashboard and design	dashboard/greenhouse-overview.json doc/design/Greenhouse_Hardware/	Grafana queries, charts and status KiCad electrical schematic

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3.2. Connections between modules



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3.3. Data Structures

Data structures used for storing and information transport

LatestReadings
Values: temperature, humidity; light / water / soil ADC
Derived: light_pct, moisture_pct, water_low, activate_pump
Driver results: dht / light / water / moisture status
Flags: air_valid, light_valid, water_valid, moist_valid
Timestamp: sampled_at_us

Watering
boot_id, cycle_id
started, occurred_at_ms
duration_ms, reason, test
Temporary struct built at a transition

Used to construct

Readings lifecycle
readSensors writes raw values + statuses
validateReadings writes validity + derived values
updateWatering reads safety inputs
displayResults and formatTelemetry read results

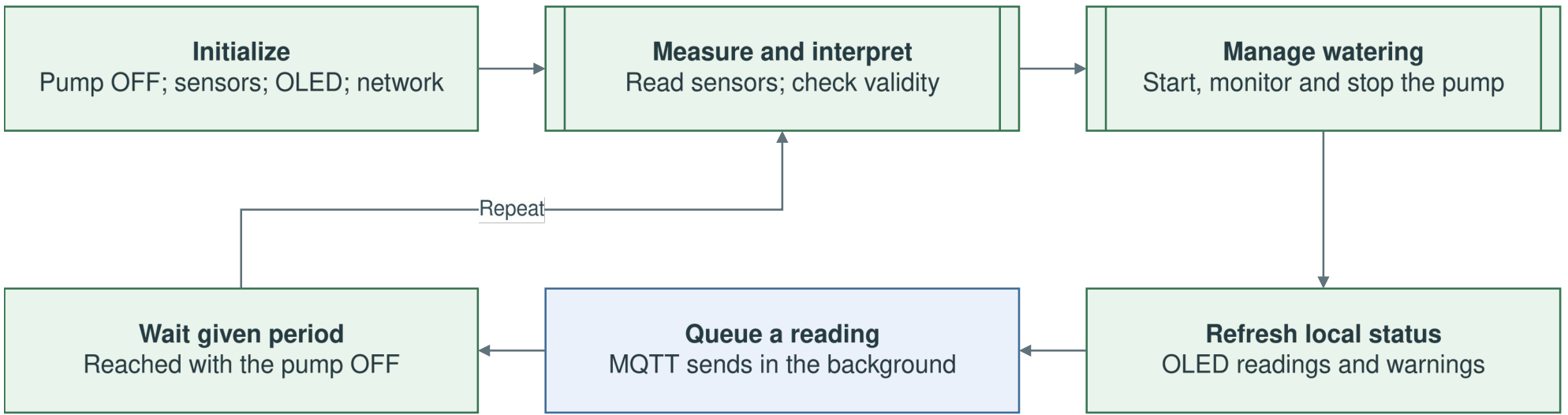
Persistent state within one boot
Global pump_on is updated by setPump()
Global watering_boot_id labels event identity
updateWatering static: cycle_id, started_at_ms, stopped_at_ms and cooling_down
No state is persisted across reset

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Firmware

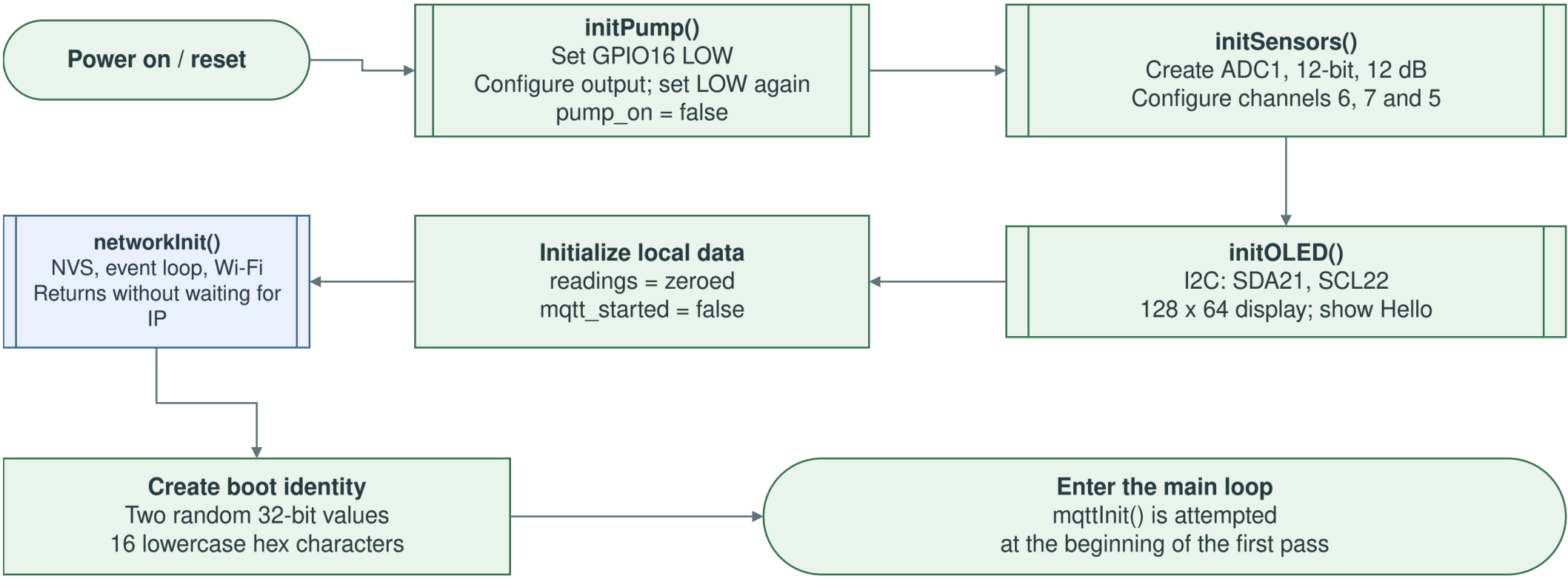
4.1. Control Loop Logic

Control Loop Logic



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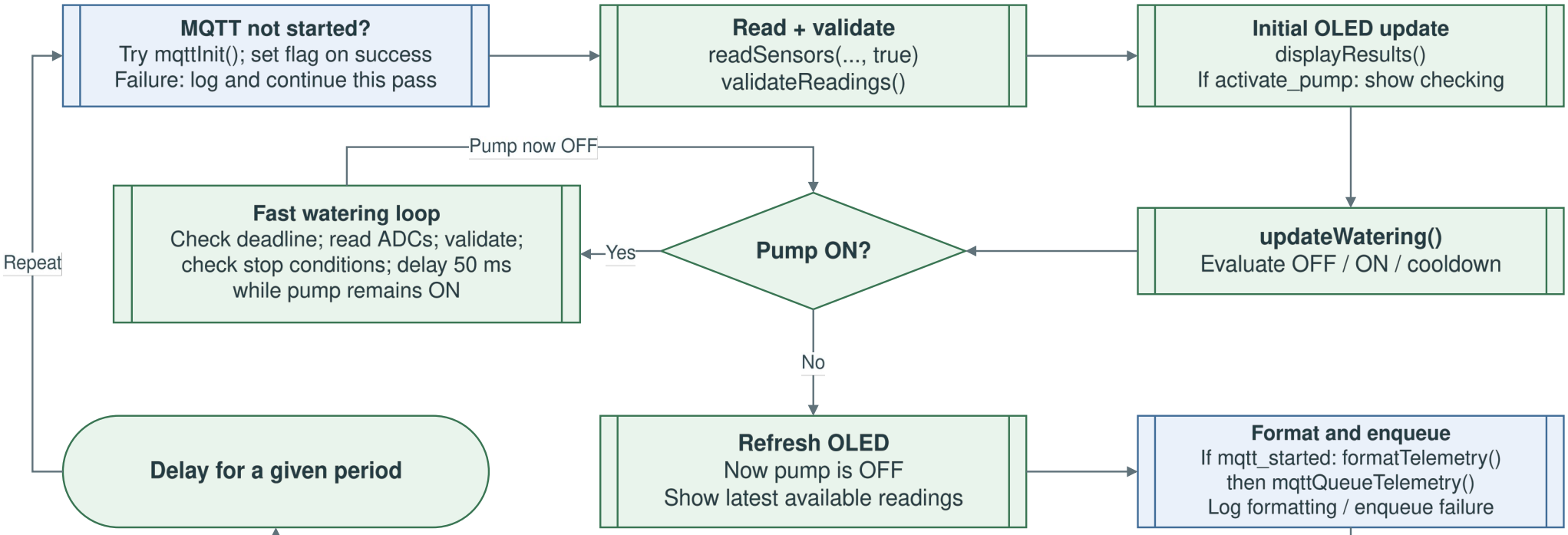
4.2. Startup Logic



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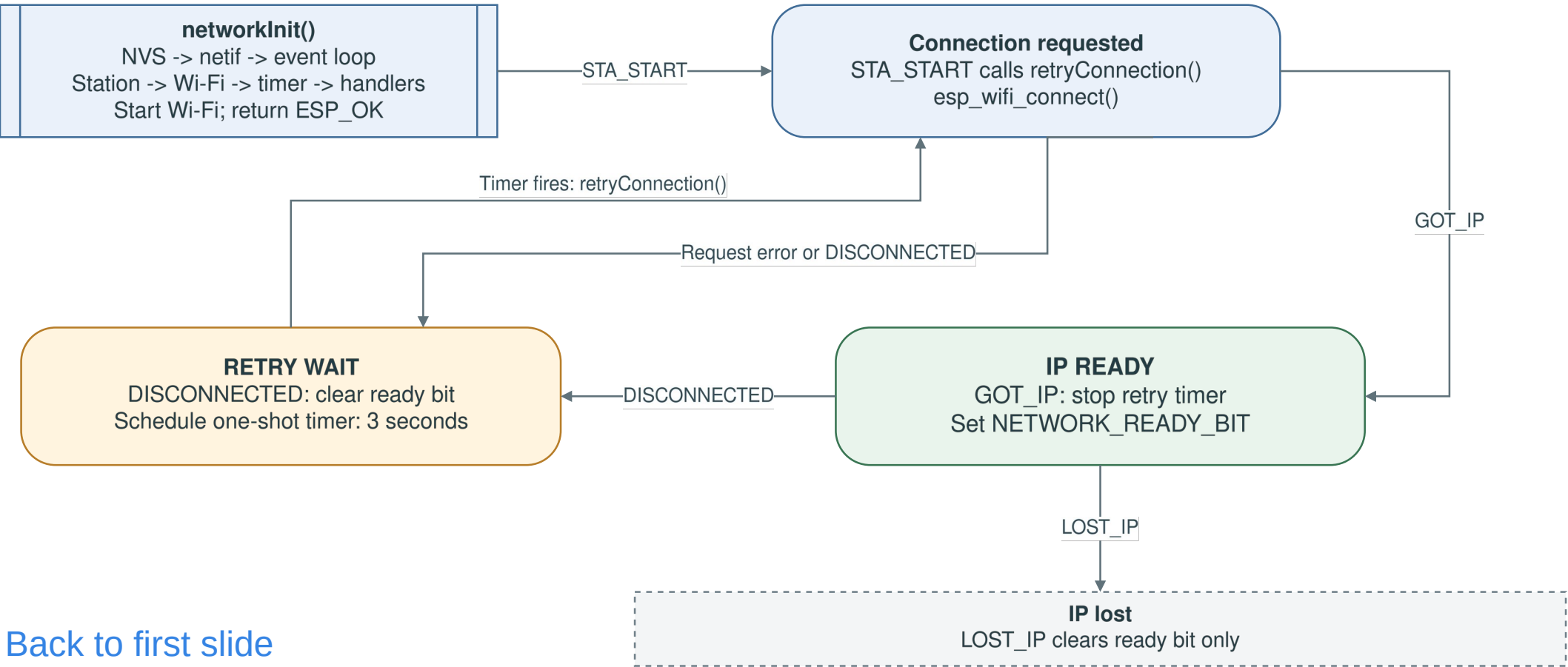
4.3. Main Loop Logic

Execution order of app_main()



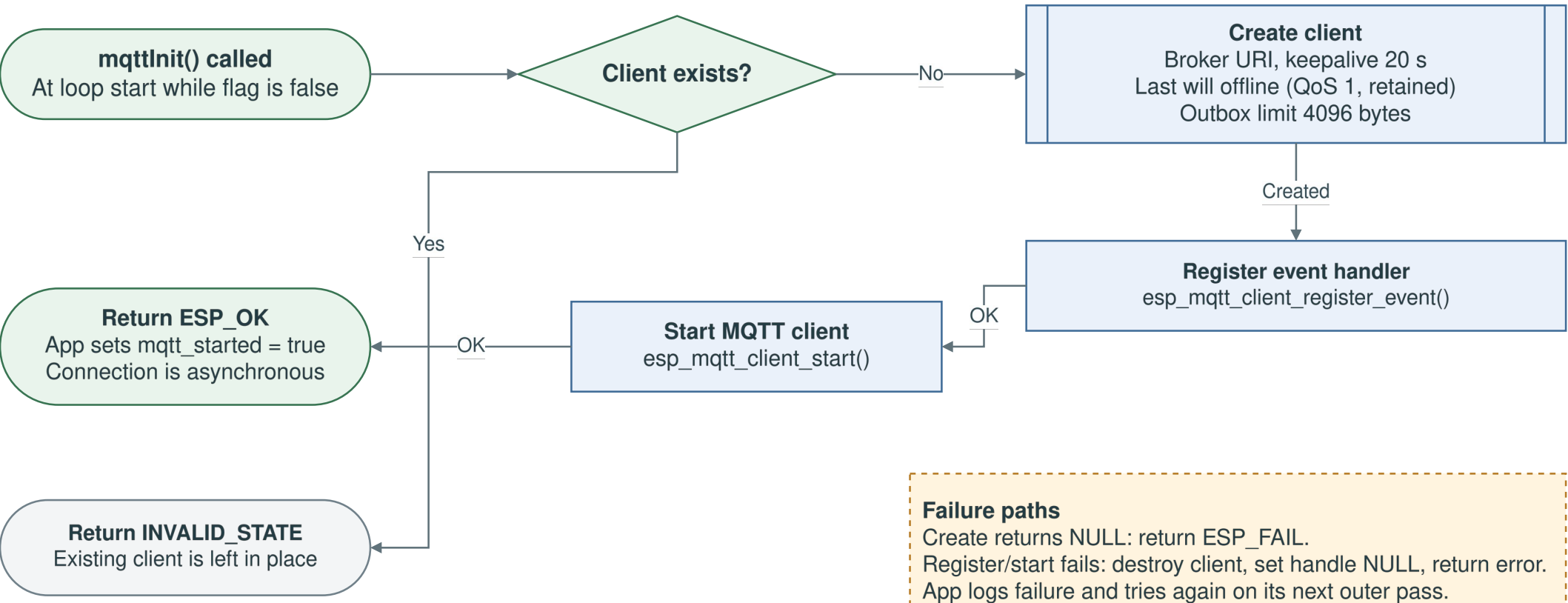
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4.4. Network logic



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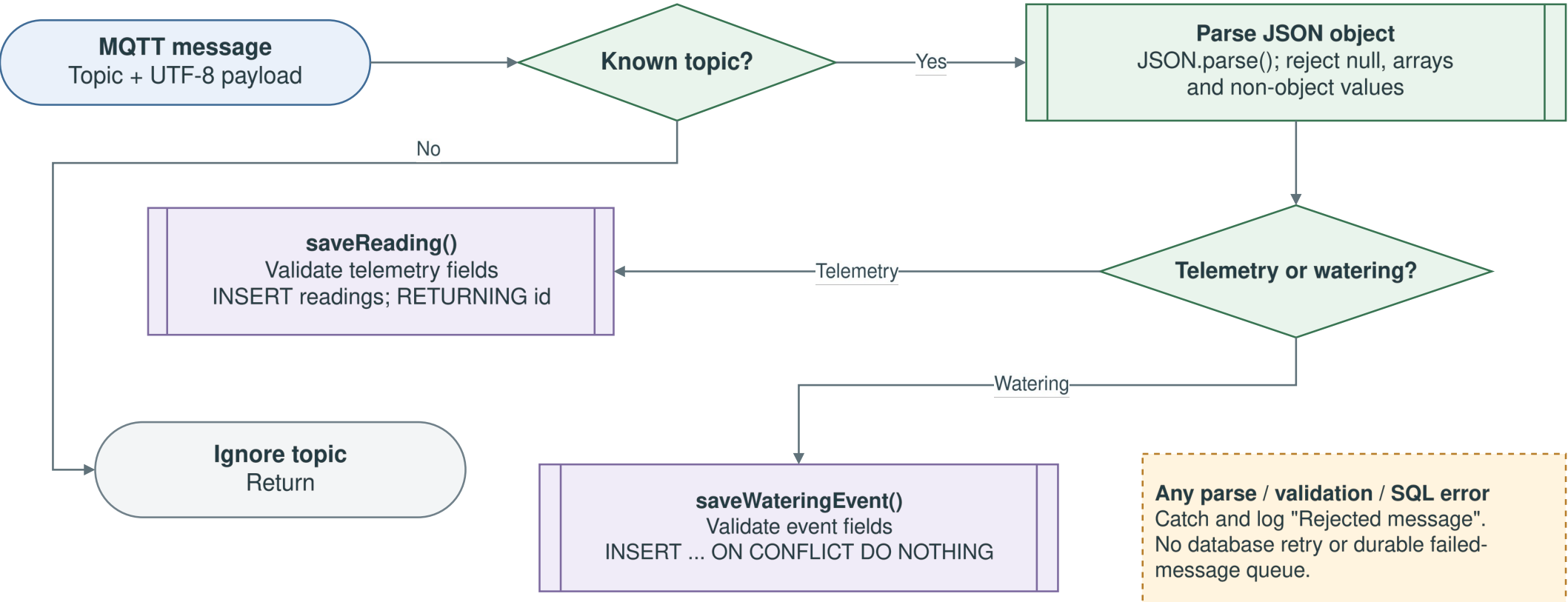
4.5. MQTT Logic



Failure paths
Create returns NULL: return ESP_FAIL.
Register/start fails: destroy client, set handle NULL, return error.
App logs failure and tries again on its next outer pass.

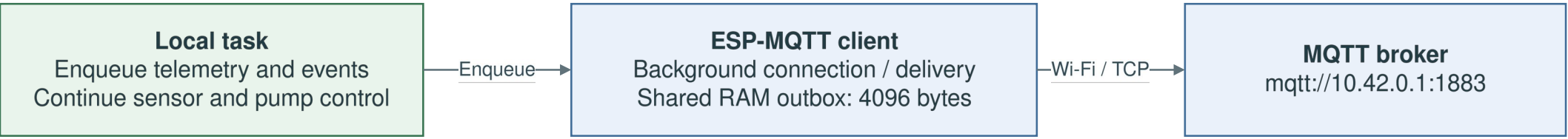
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4.6. Backend Logic



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4.7. Reporting Logic



Topic suffix under greenhouse/01/	Payload / trigger	Delivery / consumer
telemetry	JSON after an outer pass Invalid measurements become null	QoS 1; not retained Node.js -> readings
watering	JSON on pump start and stop	QoS 1; not retained Node.js -> watering_events
status	online on MQTT connect offline is the last will	QoS 1; retained Not consumed by current backend

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